Theory Assignment 1: CS2233

August 12, 2026

Total 55 Marks

Kindly adhere to the following instructions.

- Please neatly write your solutions on the answer sheet. Please clearly mention your name and roll number in the top right corner of the paper. The solutions should be self-explanatory with required mathematical arguments/proofs.
- Typed solution will not be accepted.
- Please submit the scanned copy of the solution to the google classroom portal and hard copy to the classroom.
- The submission (strict) timeline is 25th August, Tuesday, 1 PM NOON (after the class). Kindly ensure to submit both hard copy and scanned copy of the solution the given deadline. Only submitted solution be be considered for evaluation. Email solution will not be considered.
- Kindly note that the deadline is firm and no extension will be given.
- Any form of plagiarism (web/chatGPT/with peers) will be severely penalized and will result in F grade.

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1. For each of these questions, briefly explain your answer. 3 Marks
   - If I prove that an algorithm takes $O(n^2)$ worst-case time, is it possible that it takes $O(n)$ on some inputs?
   - If I prove that an algorithm takes $O(n^2)$ worst-case time, is it possible that it takes $O(n)$ on all inputs?
   - If I prove that an algorithm takes $\Theta(n^2)$ worst-case time, is it possible that it takes $O(n)$ on some inputs?
2. Let $f(n)$ and $g(n)$ be non-negative functions over non-negative inputs. Prove or disprove the following and justify your claims via formal mathematical arguments.
   8 Marks

- $f(n) + o(f(n)) = \Theta(f(n))$.
- $f(n) = O(g(n))$ implies $g(n) = \Omega(f(n))$.
- $f(n) + g(n) = \Theta(\min\{f(n), g(n)\})$
- $n! = o(n^n)$ and $n! = \omega(2^n)$

3. Is the function $\lceil \ln n \rceil!$ polynomially bounded? Is the function $\lceil \ln \ln n \rceil!$ polynomially bounded? 3 Marks

4. Solve the following recursions. Mention any assumptions you might be making. 5 Marks

   - $T(n^2) = 7T(n^2/4) + cn^2$ and $T(1) = 1$
   - $T(n) = n * T(\sqrt{n})$ and $T(2) = 4$ (Assume n to be of the form 2^{2^i})
   - $T(n) = T(n/2) + 2T(n/4) + 3n/2$ (whenever $n > 3$) and $T(1) = 0, T(2) = 2$ (assume n is a power of 2)
   - $T(n) = 4T(n/2) + n^3$ and $T(1) = 1$
   - $T(n) = T(n/2) + c \log n$ and $T(1) = \Theta(1)$ (assume n is a power of 2)

5. Arrange the following function by order of growth. That is, find an ordering among the following functions $f_1, f_2, f_3, \ldots, f_6$ such that $f_1 = \Omega(f_2), f_2 = \Omega(f_3), \ldots f_5 = \Omega(f_6)$. Partition the list into equivalence class such that functions $f(n)$ and $g(n)$ are in the same class (that is, they are of the same order), if and only if $f(n) = \Theta(g(n))$. Justify your claims.

   3 Marks

   - $\frac{n^{1.2}}{\log n}$
   - n^2
   - $n \log n$
   - 1.1^n
   - 0.9^n
   - $\log^3 n$

6. What value is returned by the following function? Express your answer as a function of n. Give the worst-case running time using big-Oh notation. 3 Marks

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function XYZ(n)
r := 0
   for i := 1 to n do
      for j := 1 to i do
         for k := j to i + j do
            for l := 1 to i + j − k do
               r := r + 1;
return(r)
```

7. Order the following functions by asymptotic growth rate, resolving ties into Θ-equivalence classes, and give a short proof for the two hardest-to-order pairs:
2 Marks

$$\log(n!), \quad n \log n, \quad n^{1.5}, \quad 2^{\sqrt{\log n}}, \quad (\log n)^{\log n}, \quad 4^n, \quad n^{3/2} \cdot \log n.$$

8. Define a family $\mathcal{H}$ of hash functions from a finite set U to a finite set B to be ϵ-universal if for all pairs of distinct elements k and l in U,

$$\Pr\{h(k) = h(l)\} \leq \epsilon,$$

where the probability is over the choice of the hash function h drawn at random from the family $\mathcal{H}$. Show that an ϵ-universal family of hash functions must have

$$\epsilon \geq \frac{1}{|B|} - \frac{1}{|U|}.$$

3 Marks

9. Let U be the set of n-tuples of values drawn from $\mathbb{Z}_p$, and let $B = \mathbb{Z}_p$, where p is prime. Define the hash function $h_b : U \to B$ for $b \in \mathbb{Z}_p$ on an input n-tuple $\langle a_0, a_1, \ldots, a_{n-1} \rangle$ from U as

$$h_b(\langle a_0, a_1, \ldots, a_{n-1} \rangle) = \left(\sum_{j=0}^{n-1} a_j b^j \right) \bmod p,$$

and let

$$\mathcal{H} = \{h_b : b \in \mathbb{Z}_p\}.$$

Argue that $\mathcal{H}$ is $\big((n-1)/p\big)$-universal according to the definition of ϵ-universal in the previous exercise. (Hint: See Exercise 31.4-4.)
4 Marks

10. Suppose n keys are inserted into a hash table of m slots using chaining, with simple uniform hashing (each key is equally likely to hash to any slot, independently). Let $\alpha = n/m$ be the load factor. 4 Marks

- Show that the expected number of keys in a slot is α.
- Prove that the expected time for an unsuccessful search is $\Theta(1 + \alpha)$.
- Prove that the expected time for a successful search is $\Theta(1 + \alpha)$, assuming a key is equally likely to be any of the n keys already in the table (assume new keys are appended to the end of a chain).

11. Perfect hashing uses a two-level scheme: keys are first hashed with a universal hash function into m slots ($m = n$, the number of keys). If n_j keys hash to slot j, a secondary hash table of size $m_j = n_j^2$ is built for that slot's keys using a hash function drawn from a universal family, so that no collisions occur within the secondary table. 5 Marks

- Using the birthday-paradox-style counting argument, show that if $m_j = n_j^2$ and the secondary hash function is drawn from a universal family on a set of size m_j, the expected number of collisions among the n_j keys is less than $1/2$. Conclude that a collision-free secondary hash function can be found by trying a constant expected number of candidates.
- Given that the level-one table hashes n keys into $m = n$ slots using a universal hash function, show that

$$\mathbb{E}\left[\sum_j n_j^2\right] < 2n.$$

  (Hint: relate $\sum_j n_j^2$ to the number of collision pairs in the level-one table, and bound this using universality.)
- Using the bound from the previous part and Markov's inequality, argue that the total expected space used by all secondary tables is $O(n)$, and that the probability the total space exceeds cn (for suitable constant c) can be made small.

12. A load-balanced hash table with chaining and "two-choice" hashing: each key is hashed by two independent hash functions h_1, h_2 into m slots, and on insertion the key is placed in whichever of the two chains is currently shorter (ties broken arbitrarily). Argue informally (a rigorous proof is not required, but a clear probabilistic argument is) why, with $n = m$ keys, the maximum chain length is $O(\log \log n)$ with high probability, in contrast to the $\Theta(\log n / \log \log n)$ maximum chain length obtained with a single random hash function. 3 Marks

13. Suppose we use open addressing with linear probing and uniform hashing on the initial probe. Show that, once slot i is examined and found occupied during an unsuccessful search, the sequence of probes is completely determined and does not depend on the key being searched for. Use this to argue why linear probing tends to develop "clusters" of occupied slots, and explain qualitatively why this makes the expected number of probes for linear probing worse than for a scheme satisfying the uniform hashing assumption, at the same load factor α. 2 Marks

14. Suppose that the initial hash location of each of n keys is chosen independently and uniformly from a hash table of size m. Let $p(n, m)$ be the probability that all n initial hash locations are distinct. 3 Marks
    Show that

$$p(n, m) \leq e^{-n(n-1)/2m}.$$

    Argue that the probability of avoiding collisions goes rapidly to zero when $n/\sqrt{m} \to \infty$.

15. Let $\mathcal{H}$ be a universal family of hash functions from U to B, with $|B| = m$. Suppose we build a chained hash table by choosing h uniformly at random from $\mathcal{H}$ (rather than assuming simple uniform hashing) and insert n arbitrary (possibly adversarially chosen) keys. 4 Marks

    - Show that the expected number of collisions (pairs of distinct keys hashing to the same slot) is at most $\binom{n}{2}/m$.
    - For any fixed stored key x, show that the expected length of the chain containing x is at most $1 + (n-1)/m = O(1 + n/m)$, matching the guarantee obtained under the simple uniform hashing assumption — even though the n keys may be adversarially chosen, as long as h is chosen randomly from $\mathcal{H}$ *after* the keys are fixed.
    - Briefly explain why this result is significant: what attack does it defeat, and against what kind of hash function would this attack succeed?